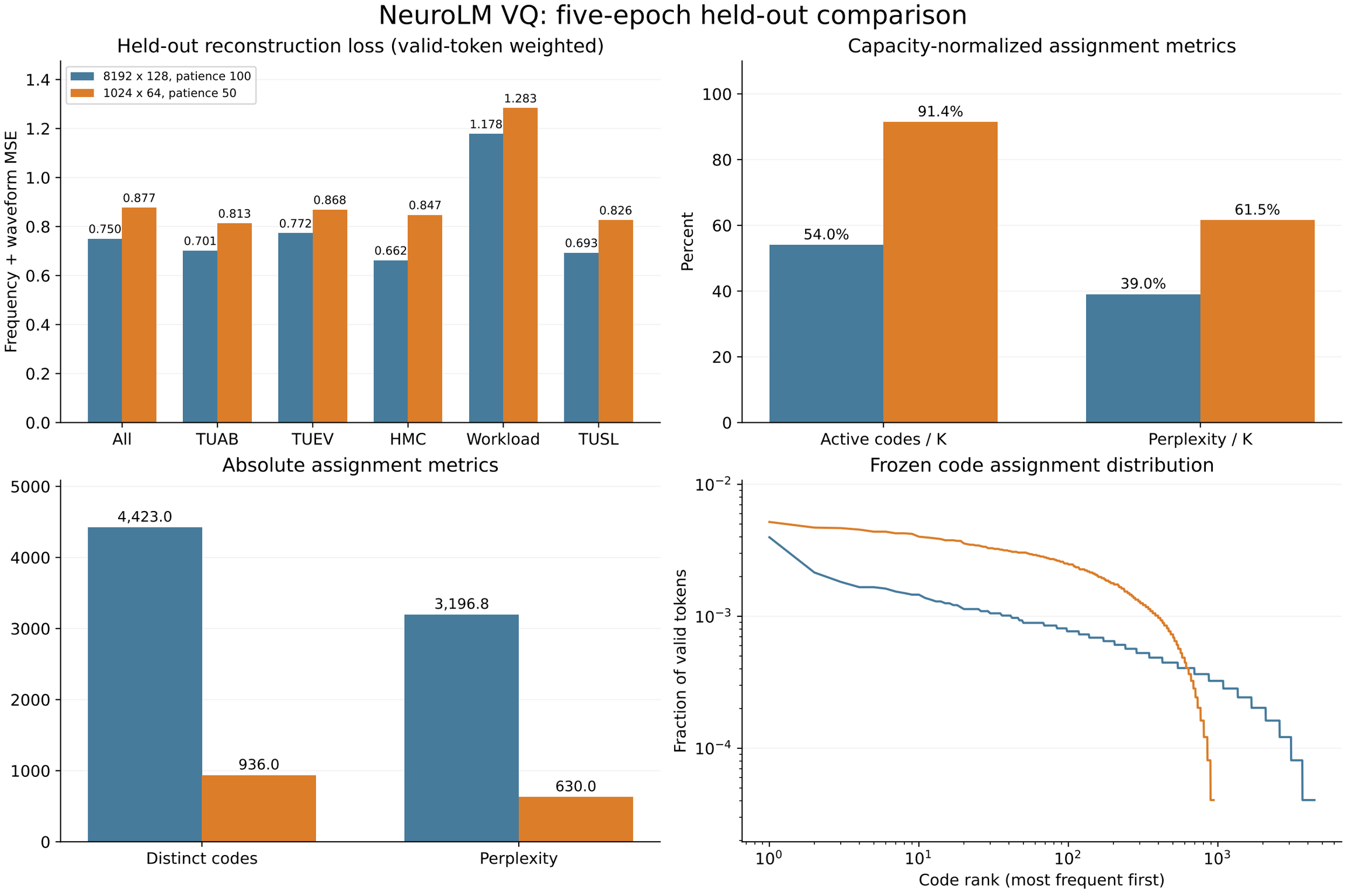
NeuroLM VQ: five-epoch held-out comparison
Held-out reconstruction loss (valid-token weighted)
8192 x 128, patience 100
1024 x 64, patience 50
Frequency + waveform MSE
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
0.750
0.877
0.701
0.813
0.772
0.868
0.662
0.847
1.178
1.283
0.693
0.826
All
TUAB
TUEV
HMC
Workload
TUSL
Capacity-normalized assignment metrics
Percent
0
20
40
60
80
100
54.0%
91.4%
39.0%
61.5%
Active codes / K
Perplexity / K
Absolute assignment metrics
0
1000
2000
3000
4000
5000
4,423.0
936.0
3,196.8
630.0
Distinct codes
Perplexity
Frozen code assignment distribution
Fraction of valid tokens
10^{-2}
10^{-3}
10^{-4}
10^{0}
10^{1}
10^{2}
10^{3}
Code rank (most frequent first)
Same 160 held-out windows (32 per dataset), 24,672 valid tokens; both VQ checkpoints frozen.
One seed; capacity, embedding dimension and reset patience change together. Not downstream task accuracy.